Chapter 5. Partitions

Divide and Conquer

Contents

5.1 Compositions **2**

5.2 Set partitions **2**

5.3 Integer partitions **4**

5.4 Selected exercises **7**

Distributions of n balls into k boxes, each identical or distinct: compositions, set partitions, integer partitions.

5.1 Compositions

Motivation. Identical balls into distinct boxes are ordered sums, counted by balls and walls.

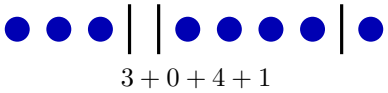
Definition 5.1 (Compositions)

A sequence $(a_1, \ldots, a_k)$ of integers with $a_i \geq 0$ and $a_1 + \cdots + a_k = n$ is a **weak composition** of n into k parts; if all $a_i > 0$, it is a **composition**.

Theorem 5.2

For positive integers n, k, the number of weak compositions of n into k parts is $\binom{n+k-1}{k-1} = \binom{n+k-1}{n}$.

Proof. n identical balls into k distinct boxes correspond bijectively to rows of n balls and $k-1$ walls: $\frac{(n+k-1)!}{n!(k-1)!}$ (Theorem 3.5). ■



$3 + 0 + 4 + 1$

Figure. *The weak composition* $(3, 0, 4, 1)$ *as* 8 *balls and* 3 *walls.*

Corollary 5.3

The number of compositions of n into k parts is $\binom{n-1}{k-1}$.

Proof [Supplement]. Subtracting 1 from every part is a bijection onto weak compositions of $n-k$ into k parts: $\binom{n-k+k-1}{k-1} = \binom{n-1}{k-1}$ (Theorem 5.2). ■

Corollary 5.4

A positive integer n has 2^{n-1} compositions (into any number of parts).

Proof.

Step 1 (summing). $\sum_{k=1}^{n} \binom{n-1}{k-1} = 2^{n-1}$ (Corollary 5.3, Theorem 4.4).

Step 2 (doubling, alternative). 1 has one composition. Each composition of n gives two of $n+1$ (add 1 to the first part, or prepend a part 1), and each composition of $n+1$ arises once (first part ≥ 2 or $= 1$). ■

(Weak compositions with unspecified number of parts are infinitely many.)

5.2 Set partitions

Motivation. Distinct balls into identical boxes split $[n]$ into blocks; recurrences follow the largest element.

Definition 5.5 (Set partitions, Stirling numbers of the second kind)

A **partition** of $[n]$ is a collection of nonempty **blocks** such that each element of $[n]$ lies in exactly one block. $S(n,k)$, the **Stirling number of the second kind**, is the number of partitions of $[n]$ into k blocks. $S(n,k) = 0$ for $n < k$; by convention $S(0,0) = 1$ and $S(n,0) = 0$ for $n > 0$.

Example 5.6

$S(n,1) = S(n,n) = 1$ $(n \geq 1)$, $S(n,n-1) = \binom{n}{2}$ $(n \geq 2$; one doubleton$)$.

Example 5.7

$S(4,2) = 7$: $\{1,2,3\}\{4\}$, $\{1,2,4\}\{3\}$, $\{1,3,4\}\{2\}$, $\{2,3,4\}\{1\}$, $\{1,2\}\{3,4\}$, $\{1,3\}\{2,4\}$, $\{1,4\}\{2,3\}$.

No closed formula exists; a single sum follows from the sieve (Chapter 7).

Theorem 5.8

For positive integers $k \leq n$:

$$S(n,k) = S(n-1,k-1) + k\,S(n-1,k). \tag{5.1}$$

Proof. If $\{n\}$ is a block, the rest is one of $S(n-1,k-1)$ partitions; otherwise removing n leaves one of $S(n-1,k)$ partitions, and n joins one of k blocks. ■

Corollary 5.9

The number of surjections $f : [n] \to [k]$ is $k!\,S(n,k)$.

Proof. Surjections correspond bijectively to partitions into k blocks (the fibres $f^{-1}(i)$) with one of $k!$ labellings by $[k]$. ■

Corollary 5.10

For every real x and nonnegative integer n:

$$x^n = \sum_{k=0}^{n} S(n,k)(x)_k, \tag{5.2}$$

where $(x)_k = x(x-1)\cdots(x-k+1)$.

Proof. $n = 0$: both sides 1. Both sides are polynomials of degree n, so check $x \in \mathbb{N}$. Then x^n counts functions $[n] \to [x]$ (Theorem 3.6); grouped by image of size k, there are $\binom{x}{k} \cdot k!\,S(n,k) = (x)_k S(n,k)$ (Corollary 5.9). ■

Definition 5.11 (Bell numbers)

$B(n)$, the n-th **Bell number**, is the number of all partitions of $[n]$; $B(0) = 1$. So $B(n) = \sum_{i=0}^{n} S(n,i)$.

Theorem 5.12

For every nonnegative integer n:

$$B(n+1) = \sum_{i=0}^{n} \binom{n}{i} B(i). \tag{5.3}$$

Proof. If i elements lie outside the block of $n+1$: $\binom{n}{i}$ choices, $B(i)$ partitions of them. Sum over i. ∎

5.3 Integer partitions

Motivation. With nothing distinguishable only box sizes remain; Ferrers diagrams turn identities into pictures.

Definition 5.13 (Integer partitions)

If $a_1 \geq a_2 \geq \cdots \geq a_k \geq 1$ are integers with $a_1 + \cdots + a_k = n$, then $(a_1, \ldots, a_k)$ is a **partition** of the integer n. $p(n)$ denotes the number of partitions of n, and $p_k(n)$ the number with exactly k parts.

Example 5.14

$p(5) = 7$: $(5), (4,1), (3,2), (3,1,1), (2,2,1), (2,1,1,1), (1,1,1,1,1)$.

There is no simple recurrence (but see the Notes). Asymptotically

$$p(n) \sim \frac{1}{4n\sqrt{3}} \exp\left(\pi\sqrt{\frac{2n}{3}}\right), \tag{5.4}$$

faster than any polynomial and slower than any c^n, $c > 1$.

The **Ferrers shape** of $(x_1, \ldots, x_k)$ has x_i left-justified boxes in row i. Reflecting in the main diagonal gives the **conjugate**; that of $(4,2,1)$ is $(3,2,1,1)$.

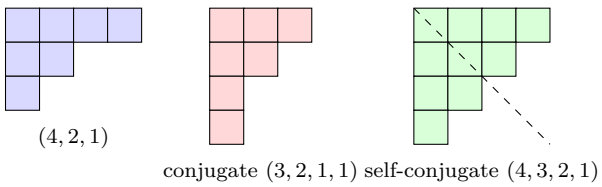


Figure. *Ferrers shapes; conjugation reflects in the diagonal (dashed).*

Definition 5.15 (Self-conjugate)

A partition is **self-conjugate** if it equals its conjugate.

Example 5.16

$(4,3,2,1)$, $(5,1,1,1,1)$ and $(4,2,1,1)$ are self-conjugate.

Theorem 5.17

The number of partitions of n into at most k parts equals the number of partitions of n into parts not larger than k.

Proof. Conjugation swaps shapes with at most k rows and at most k columns. ■

(For $n = 5$, $k = 2$: $(5), (4,1), (3,2)$ versus $(2,2,1), (2,1,1,1), (1^5)$.)

Theorem 5.18

The number of partitions of n into distinct odd parts equals the number of self-conjugate partitions of n.

Proof.

Step 1 (the map). For self-conjugate π, the first row and column form the **hook** of $2\pi_1 - 1$ boxes; repeating, the i-th hook has $2\pi_i - (2i-1)$ boxes. $f(\pi) = (2\pi_1 - 1, 2\pi_2 - 3, \dots)$ has odd, strictly decreasing parts: $(2\pi_i - 2i + 1) - (2\pi_{i+1} - 2i - 1) = 2(\pi_i - \pi_{i+1}) + 2 \geq 2$.

Step 2 (inverse). For $\alpha = (2a_1 - 1, \dots, 2a_u - (2u-1))$, $f(\pi) = \alpha$ forces a_i boxes in row and column i of the i-th hook, a unique self-conjugate shape with image α. ■

Example 5.19

For self-conjugate $\pi = (6,6,4,3,2,2)$, $f(\pi) = (11,9,3)$.

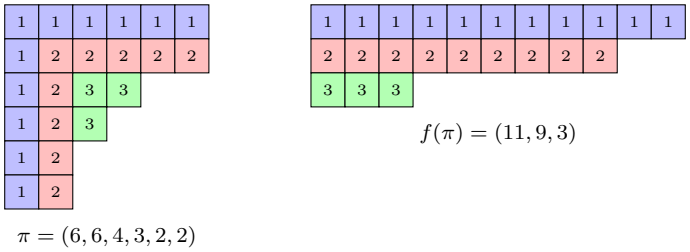


Figure. *Theorem 5.18: hook i becomes row i.*

Theorem 5.20

Let $q(n)$ be the number of partitions of n with all parts at least 2. Then $q(n) = p(n) - p(n-1)$ for $n \geq 2$.

Proof. Appending a part 1 is a bijection from partitions of $n-1$ onto partitions of n with a part 1; the rest are counted by $q(n)$. ■

Type of a set partition. Its block sizes, sorted decreasingly, form a partition of n, the **type**.

Example 5.21

$\{1,5,6\}, \{2,7\}, \{3,9\}, \{4,8\}, \{10\}$ has type $(3,2,2,2,1)$.

Theorem 5.22

Let $a = (a_1, \dots, a_k)$ be a partition of n and m_i the multiplicity of i as a part of a. The

number of set partitions of $[n]$ of type a is

$$P_a = \frac{\binom{n}{a_1,\dots,a_k}}{\prod_{i\ge 1} m_i!}.$$

Proof. Order a_j balls of colour j ($j \in [k]$) in $\binom{n}{a_1,\dots,a_k}$ ways (Theorem 3.5); same-coloured positions form the blocks, giving every partition of type a. Two colourings give the same partition iff they differ by permuting equal-size colour classes, so each arises $\prod_i m_i!$ times. ■

Example 5.23

There are $\binom{10}{3,2,2,2,1}/(1!\,3!\,1!) = 12600$ set partitions of $[10]$ of type $(3,2,2,2,1)$.

Example 5.24

The partitions of $[2n]$ into blocks of size 2 (perfect matchings) number

$$\frac{\binom{2n}{2,2,\dots,2}}{n!} = \frac{(2n)!}{2\cdot 4\cdots 2n} = \prod_{i=1}^{n}(2i-1) = (2n-1)!!.$$

	Balls	Boxes	Number
Surjections	n distinct	k distinct	$k!\,S(n,k)$
	n distinct	any number, distinct	$\sum_{i=1}^{n} i!\,S(n,i)$
Compositions	n identical	k distinct	$\binom{n-1}{k-1}$
	n identical	any number, distinct	2^{n-1}
Set partitions	n distinct	k identical	$S(n,k)$
	n distinct	any number, identical	$B(n)$
Integer partitions	n identical	k identical	$p_k(n)$
	n identical	any number, identical	$p(n)$
empty boxes allowed (k boxes fixed):			
Functions	n distinct	k distinct	k^n
Weak compositions	n identical	k distinct	$\binom{n+k-1}{k-1}$
Set partitions	n distinct	k identical	$\sum_{i=1}^{k} S(n,i)$
Integer partitions	n identical	k identical	$\sum_{i=1}^{k} p_i(n)$

Notes

With the **pentagonal numbers** $k(3k-1)/2$, $k \in \mathbb{Z}$ (that is $0, 1, 2, 5, 7, 12, \dots$), MacMahon's recurrence reads

$$p(n) = p(n-1) + p(n-2) - p(n-5) - p(n-7) + \cdots, \tag{5.5}$$

the i-th sign being $(-1)^{\lfloor i/2 \rfloor}$; e.g. $p(8) = 15 + 11 - 3 - 1 = 22$. Euler: partitions into an even and an odd number of distinct parts are equinumerous unless n is pentagonal (difference one).

5.4 Selected exercises

Motivation. Book exercises on Bell and Stirling numbers.

Exercise 5.2 (Bell numbers versus factorials)

Prove that $B(n) < n!$ for $n \geq 3$.

Solution.

Step 1 (an injection). For a partition π, let σ_π act on each block $\{y_1 < \cdots < y_r\}$ as the cycle $y_1 \mapsto y_2 \mapsto \cdots \mapsto y_r \mapsto y_1$. Its cycles are the blocks, so $\pi \mapsto \sigma_\pi$ is injective: $B(n) \leq n!$.

Step 2 (not surjective). For $n \geq 3$, $\tau = (1\,3\,2)$ (fixing $x \geq 4$) is not some σ_π: π would have block $\{1, 2, 3\}$, but $\sigma_\pi(1) = 2 \neq 3$. So $B(n) \leq n! - 1$. ■

Exercise 5.16 (An alternating sum of Stirling numbers)

Prove that for all positive integers n,

$$\sum_{k=1}^{n} (-1)^k k!\, S(n,k) = (-1)^n.$$

Explain the meaning of this identity in words.

Solution. Put $x = -1$ in Corollary 5.10, with

$$(-1)_k = (-1)(-2)\cdots(-k) = (-1)^k k!$$

and $S(n, 0) = 0$.

Meaning: $k!\, S(n,k)$ counts *ordered* partitions into k blocks (Corollary 5.9); those with an even and an odd number of blocks differ by one, the even ones in excess iff n is even. ■

Exercise 5.34 (Partitions without singleton blocks)

Let $F(n)$ be the number of partitions of $[n]$ with no singleton block. Prove that $B(n) = F(n) + F(n+1)$ for $n \geq 1$, by a bijection.

Solution. Partitions of $[n]$ without singletons ($\mathcal{P}_0$) number $F(n)$; we biject those with a singleton ($\mathcal{P}_1$) onto the set $\mathcal{F}_{n+1}$ of partitions of $[n+1]$ without singletons.

Step 1 (the map). For $\pi \in \mathcal{P}_1$ let $S \neq \emptyset$ be the union of its singletons; $\Phi(\pi)$ replaces them by the block $S \cup \{n+1\}$. So $\Phi(\pi) \in \mathcal{F}_{n+1}$.

Step 2 (the inverse). For $\rho \in \mathcal{F}_{n+1}$ the block of $n+1$ is $T \cup \{n+1\}$, $T \neq \emptyset$; $\Psi(\rho)$ splits T into singletons. $\Psi(\rho) \in \mathcal{P}_1$.

Step 3 (inverse maps). The singletons of $\Psi(\rho)$ are exactly those from T (other blocks have ≥ 2 elements), so $\Phi\Psi = \mathrm{id}$, and $\Psi\Phi = \mathrm{id}$ by construction. Hence $B(n) = F(n) + F(n+1)$. ■